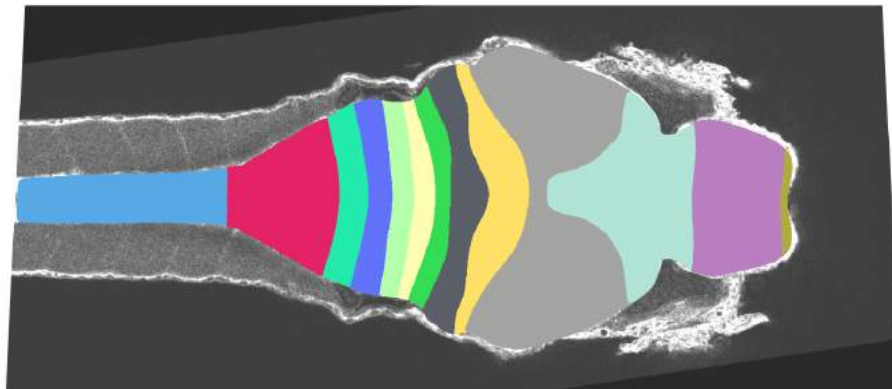
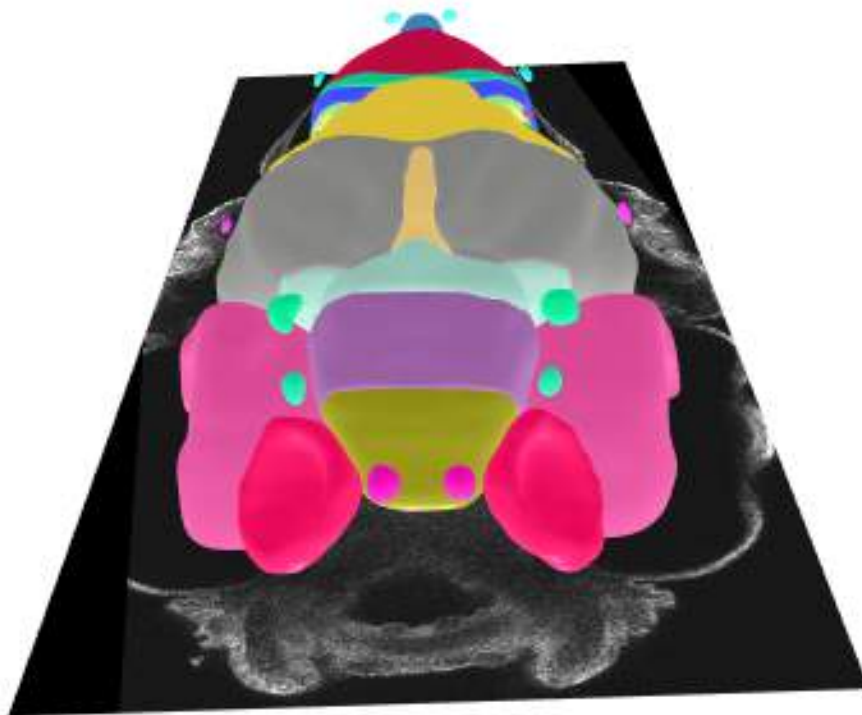


Vis-Explorer

Vis-Explorer is web-based framework for the visualization, exploration and analysis of zebra-fish brain. The overall goal is to create a realistic multi-scale circuit model of the larval zebrafish brain from the nanoscale at the synaptic level, to the micro-scale describing local circuits and to the macro-scale brain-wide activity patterns distributed across many regions. The idea is to develop generic explorer that can also be used to explore data-sets from different domains.

Website Link: <https://visual.zib.de/2018/zebrafish/visexplorer/>

Softwares: [BabylonJs](#) , [GoldenLayout](#) and [D3js](#)



Layout

Vis-Explorer consists of multi-window layout manager having three building blocks i.e. row, column and the stack view as shown in figure 1. These windows can be nested infinitely within each other. The user can drag the window to any location by selecting the header of the window. We are basically using layout manager from [GoldenLayout](#) in our platform.



Figure 1: Row, Column, Stack and Combined Layout

Visualization Plugins

Plugins are the core of Vis-explorer. The plugin mechanism enables researcher to add new features easily in the platform. There are list of plugins that are available in the Vis-explorer.

- List-Viewer
- 2D-Viewer
- 3D-Viewer
- Tree-Viewer
- Cell-Viewer
- Spatial Query Plugin

Most of the plugins are installed by default. One can install the new plugins by clicking the viewer button in menu bar as shown in figure 2. The list of viewers will appear and user can then install the respective plugins by clicking on them. Currently, one can install only single copy of plugins in the application. In next version, user will be able to add multiple plugins and can also configure them.

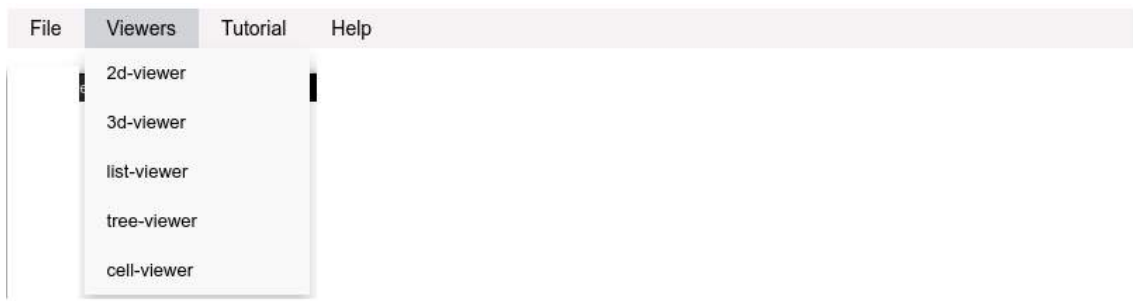


Figure 2: Install Plugin

List-Viewer

It is a helper viewer as shown in figure 3 to load the anatomical data in 2D and 3D viewer respectively. It offers below functionalities but we will explain them all in 2D and 3D viewer section with examples.

- Search anatomical regions.
- Change version space.
- Hide/ Invert regions.
- Change color, opacity and enable wire-frame view.
- Display statistic information about regions.

2D-Viewer

2D-Viewer is to map, annotate and visualize the anatomical regions in 2-dimensional reference zebra-fish brain. In general, one can map the anatomical regions to any other data-set. It is created using [BabylonJs](#) along with custom defined functionalities. Following are the list of functionalities offered by the 2D-Viewer:

- **Select/Deselect the regions:** In order to load the anatomical regions in 2D and 3D viewers, researchers need to mark the checkbox in front of anatomical regions in list-viewer as shown in figure 3. De-selecting the checkbox will remove the region from the viewer.

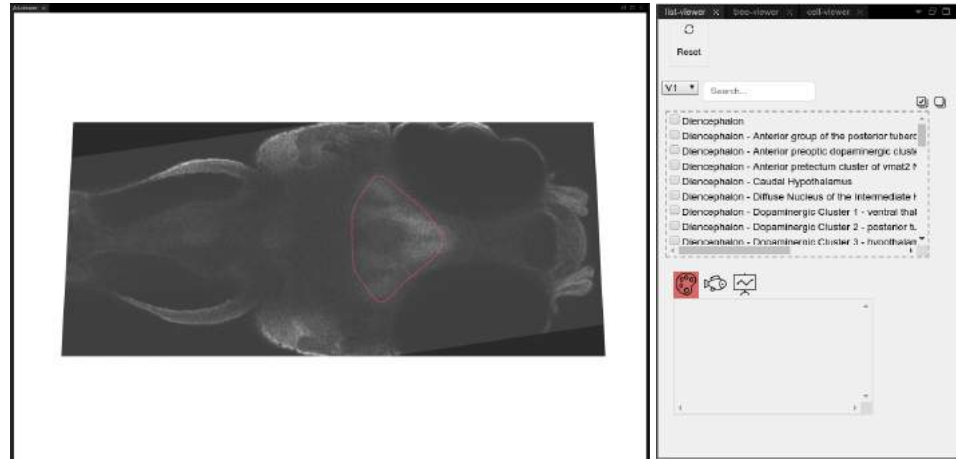


Figure 3: 2D-Viewer

- **Prev/Next Slice:** Click the orange color setting bar as shown in figure 4. Next and previous buttons will appear on the left and are used for loading next and previous slice in the viewer. User can visualize all the whole slices by clicking play button.

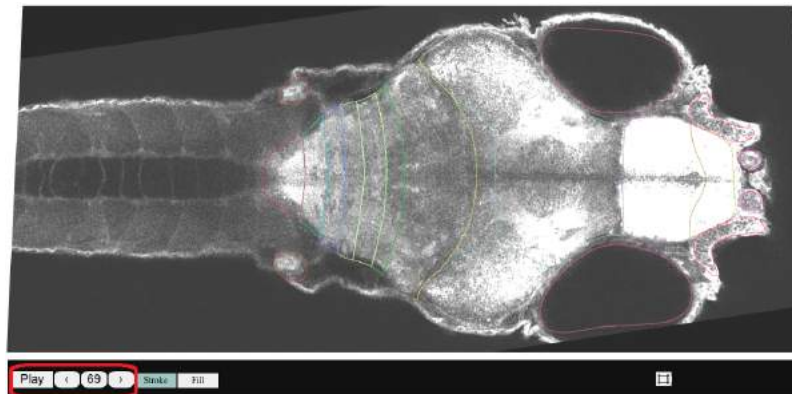


Figure 4: Play, Next and Previous

- **Outlined/Filled Anatomical Regions:** Anatomical regions can be displayed as outlined or filled contours as shown in figure 5. The default is outlined.
- **360 degree Rotation:** The Slice can be rotated in any direction as shown in figure 6
- **Zoom-in/ Zoom Out:** Slice can be zoomed in and out using the mouse wheel as shown in figure 7

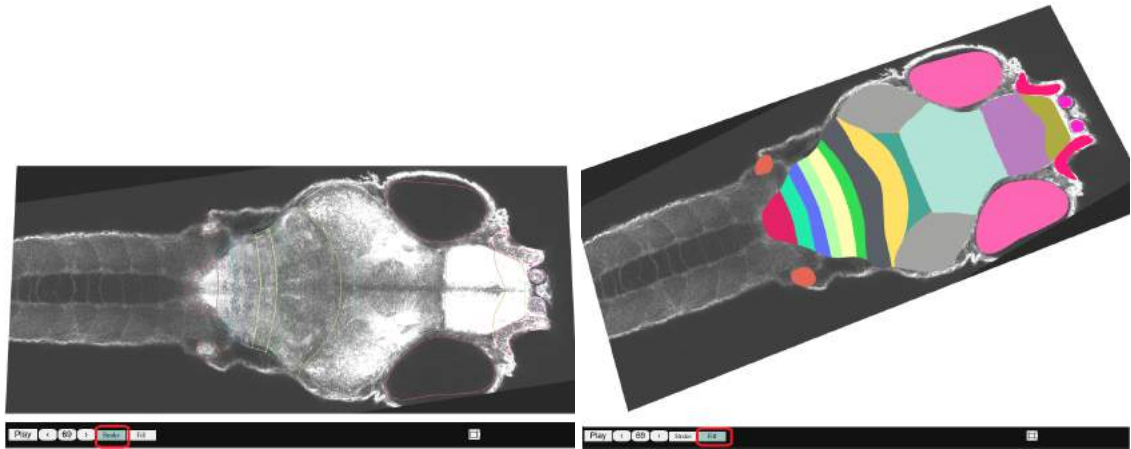


Figure 5: Outlined and Filled Anatomical Regions in 2D-Viewer

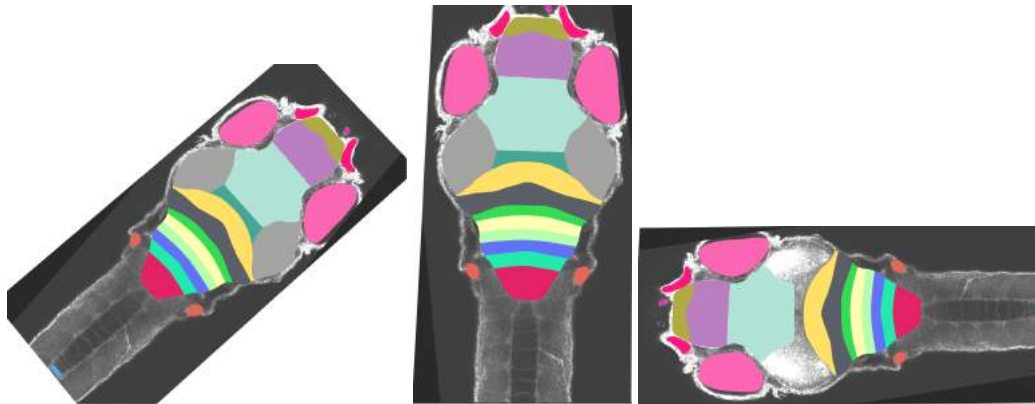


Figure 6: Rotation in 2D-Viewer

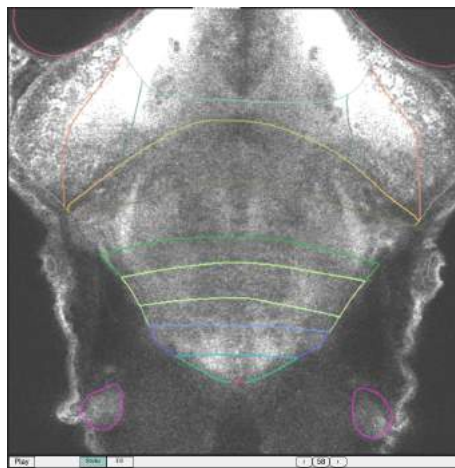


Figure 7: Zoom-in View

3D-Viewer

3D-viewer is to visualize anatomical regions in 3-dimensional space. The regions are loaded in a similar fashion like 2D-viewer. It also offers zoom-in, zoom-out and rotation of object similar

to 2D-viewer.

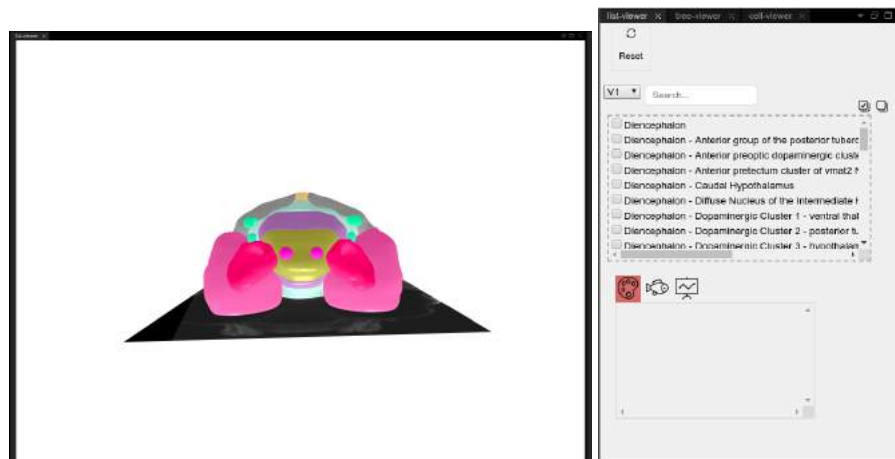


Figure 8: 3D-Viewer

It offers following extra functionalities:

- **Slice Mapping in 3D:** The slice can be moved up and down in the Z direction using previous and next button as shown in figure 9.

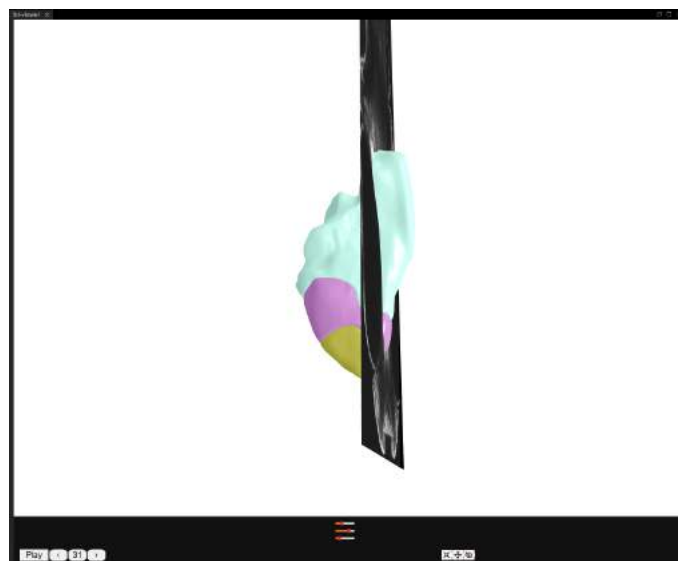


Figure 9: Slice Motion

- **Change Color and Opacity :** Click the name of the anatomical region in the list viewer. The opacity slider and change color button appears in the tabs as shown in figure 10

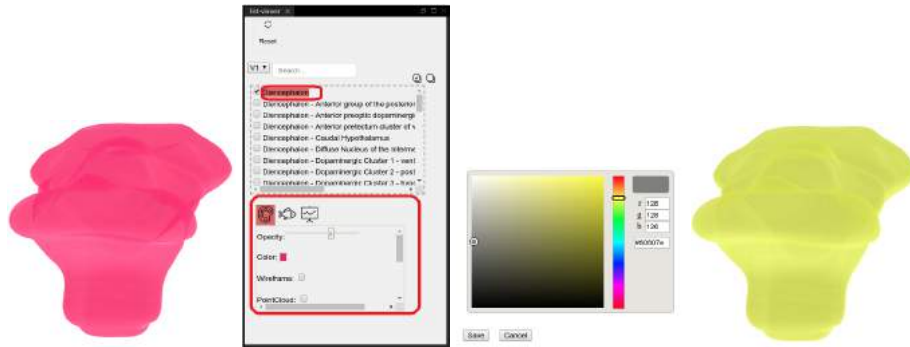


Figure 10: Change Color Pop-up

- **Wire-frame:** It is a technique to represent the three-dimensional shape of solid objects using metal wire. User should click the region in the list viewer and then the wire-frame checkbox in the tabs window as shown in figure 11.

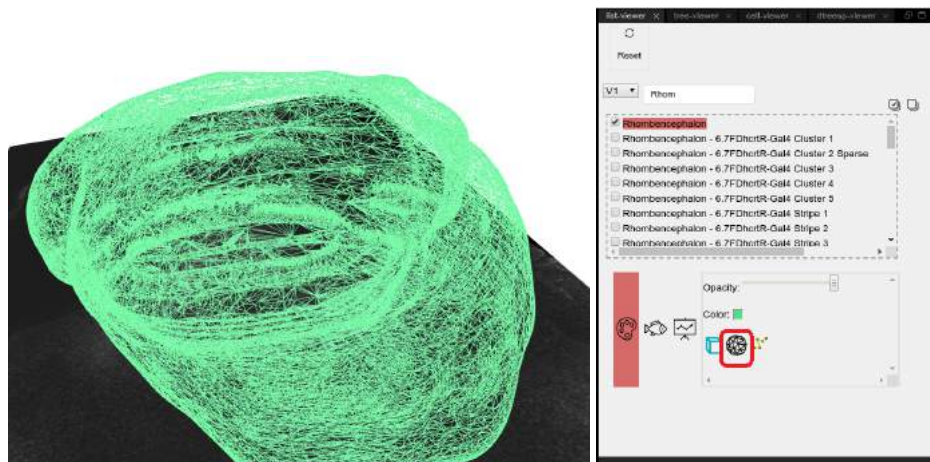


Figure 11: Wire-frame View

- **Point Cloud :** It is a technique to represent the three-dimensional shape of solid objects as data points. User should click the region in the list viewer and the point cloud checkbox in the tabs as shown in figure 12.



Figure 12: Point Cloud View

- **Drag and Drop 3D Objects :** User can drag and drop anatomical regions anywhere in the viewer by enabling the drag and drop functionality from setting tab as shown in fig 13. The user can disable it by clicking the same button again.

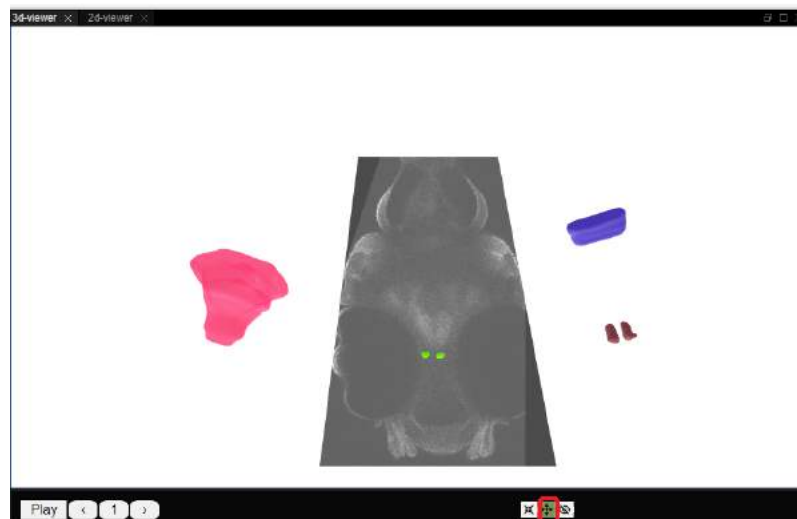


Figure 13: Drag and Drop Anatomical Regions

- **Collapse 3D Objects:** User can move back dragged anatomical regions to their original locations. This is achieved by clicking the “collapse button”, encircled red in the in fig 14.

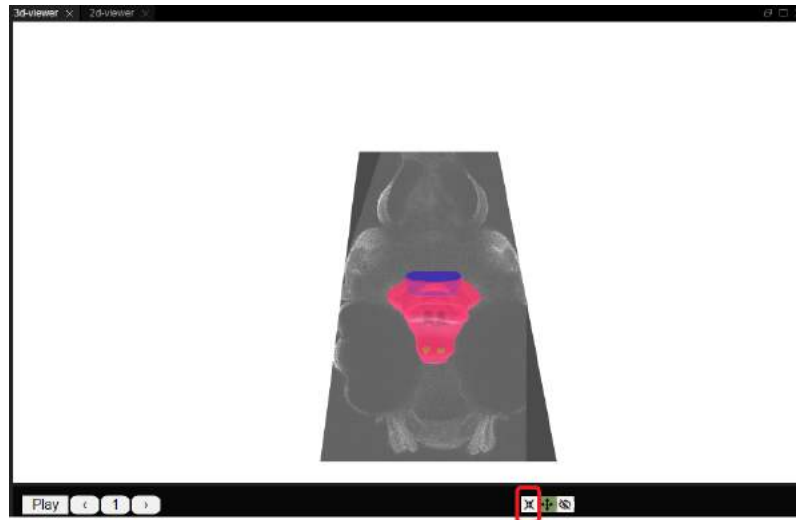


Figure 14: Collapsed Anatomical Regions

Tree-Viewer

It is used to visualize the semantic information like ontology of zebra-fish. Currently, we presented MECE hierarchy defined by [Owen Rendlet](#). Each node in the ontological tree, representing an anatomical region or a group of anatomical regions, is displayed by an orange circle. The area inside the orange circle is colored white when the hierarchy below the node is expanded or the node is a leaf node; it is colored blue when there exists a hierarchy below, which is not expanded. For example, the inner circle of the node representing the hindbrain is blue, as it is an intermediate node in the ontological tree and its sub-hierarchy is not expanded. Whereas the inner circle of the node representing the spinal cord is white, as it is a leaf node in the ontology currently used. The tree is created using [D3js](#).

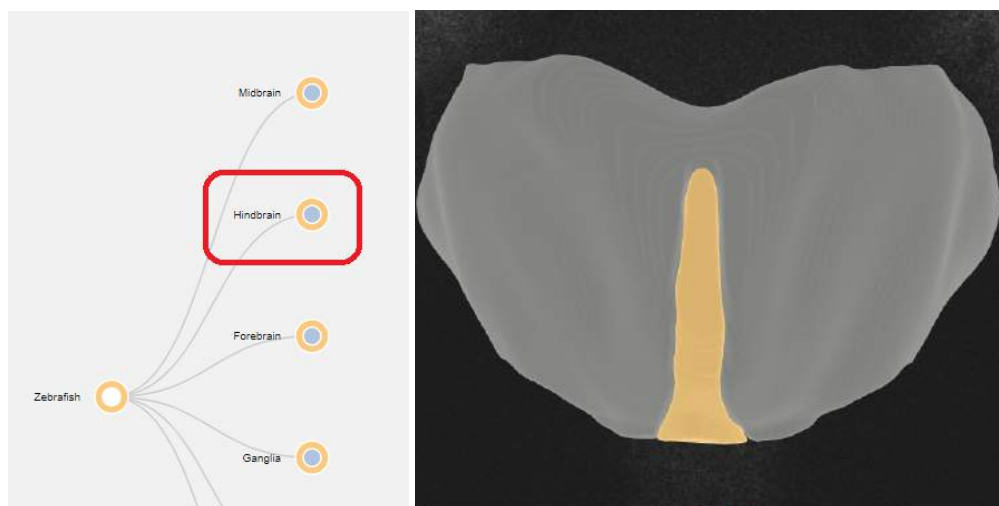


Figure 15: Hierarchical-Viewer

Following are the list of functionalities:

- **Toggle Hierarchies:** Navigation through hierarchies is possible by clicking on circular node as shown in figure 14.

- **Load/ Remove Anatomical Regions :**User should click the outer circle of anatomical region, the corresponding region and its hierarchical structures will be loaded and outer circle will turn to green. The regions can be removed by clicking the outer circle again.
- **Change Color:** Color can be changed by clicking the name of the anatomical region. The change color pop-up will appear on screen.

Cell-Viewer

For research it is important to display groups of cells in the standard brain to understand their spatial organization. We incorporate the cellular information provided by [Armin Bahl](#).

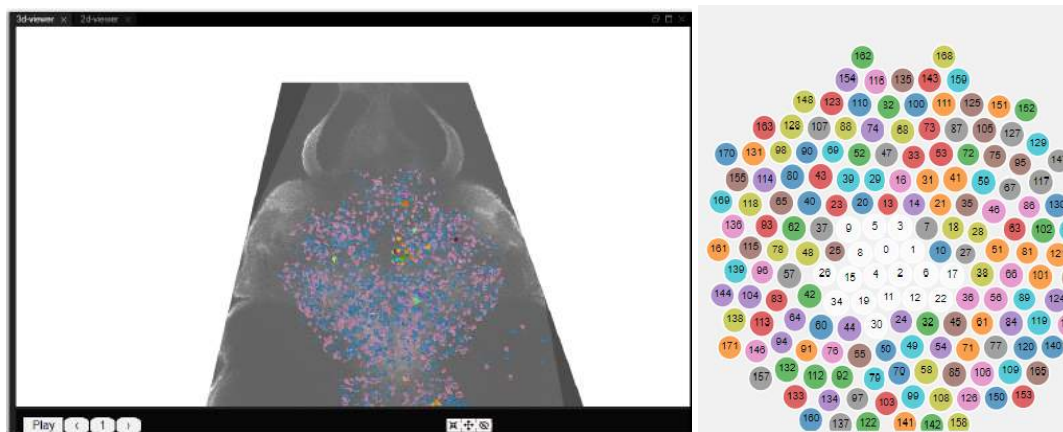


Figure 16: Cell-Viewer

- **Load/ Remove Cells :** User can display the cells in the 2D- and 3D-viewer by clicking on individual cells in the cell viewer. As an example, we have displayed the cells with numbers 0, 1, 8 and so on. For cells displayed in the 2D- and 3D-viewer the color in the cell viewer turns to white (and it is toggled back to the original color, after deselection).

Query-Viewer

Query-viewer enables researchers to depict spatial relation between anatomical regions like containment, overlap and touching etc. User needs to install query plugin as defined in [plugins](#) section. Currently, the spatial information is only available for version 1 regions only. Following are the steps to perform spatial query:

- Select anatomical region like Diencephalon as shown in figure 17.

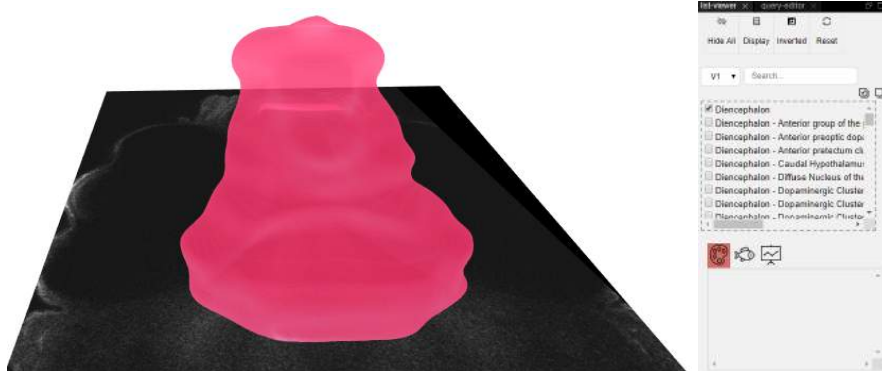


Figure 17: Select Anatomical region

- Click on one of the example query like "Within Diencephalon" in query editor plugin. The query will appear in the text editor. Click on execute query in the bottom right as shown in figure 18.



Figure 18: Select Query

- List viewer will be updated with the anatomical regions which are contained within Diencephalon. User can then load all the regions using check all button as shown in figure 19.



Figure 19: Visualize Query Results